Augmented Lagrangian Methods

Constrained optimization with smooth functions.
Consider the equality constrained case

$$\min f(x) \quad \text{s.t.} \quad c_i(x) = 0, \quad i \in \mathcal{E}.$$

Instead solve a Lagrangian form with quadratic penalty

$$\min L_A(x, \lambda; \mu) = f(x) - \sum_{i \in \mathcal{E}} \lambda_i c_i(x) + \frac{1}{2}\mu \sum_{i \in \mathcal{E}} c_i^2(x)$$

$$\min L_A(x, \lambda; \mu) = f(x) - \lambda^T c(x) + \frac{1}{2}\mu \| c(x) \|^2$$

By (eventually) choosing μ large enough $c(x) \to 0$.
Consider an iterative approach to finding a stationary point of L_A:

$$\nabla L_A(x_k, \lambda^k; \mu_k) = \nabla f(x_k) - \sum_{i \in \mathcal{E}} [\lambda_i^k - \mu_k c_i(x_k)] \nabla c_i(x_k)$$

Notice that if $\nabla L_A \approx 0$, then $\lambda_i^k - \mu_k c_i(x_k)$ may be a good estimate of λ_i^*. What might this mean?

$$\lambda_i^k - \mu_k c_i(x_k) \approx \lambda_i^* \implies c_i(x_k) \approx \frac{\lambda_i^k - \lambda_i^*}{\mu_k}$$

Message: $c_i(x_k) \to 0$ as $\lambda_i^k \to \lambda_i^*$ <u>and</u> as $\mu_k \to \infty$.

EQUALITY CONSTRAINED AL ALGORITHM

- Given : $x_0 \in \mathbb{R}^n$, $\lambda^0 \in \mathbb{R}^m$
- Set : $\mu_0 > 0$, $t_0 > 0$, $\rho \in (0,1)$, $k=0$
- while stopping criterion not met
  - Solve the unconstrained problem $\min_x L_A(x, \lambda^k; \mu_k)$ at fixed λ^k and μ_k. Terminate when either $\|\nabla L_A\| < t_k$ or iter = 2n. Return $\bar{x}$, iter, $c(\bar{x})$
  - Updates (some suggestions)
    - $\lambda^{k+1} = \lambda^k - \mu_k c(\bar{x})$ ← λ^* estimate
    - $\mu_{k+1} = (1 + 10e^{-\text{iter}/n})\mu_k$ ← μ increase ($\mu_{k+1} = 1.5\mu_k$)
    - $t_{k+1} = \rho t_k$ ← decrease t ($\rho = 1/3$)
    - $x_{k+1} = \bar{x}$
    - $k \leftarrow k+1$

end while

Inequalities?

$$\begin{aligned} \min\ & f(x) \\ \text{s.t. } & c_i(x) \geq 0,\ i \in \mathcal{I} \\ & c_i(x) = 0,\ i \in \mathcal{E} \\ & x \in \mathbb{R}^n \end{aligned} \quad \Rightarrow \quad \begin{aligned} \min\ & f(x) \\ \text{s.t. } & c_i(x) - y_i^2 = 0,\ i \in \mathcal{I} \\ & c_i(x) = 0,\ i \in \mathcal{E} \\ & x \in \mathbb{R}^n \\ & y \in \mathbb{R}^{|\mathcal{I}|} \end{aligned}$$

$$\min \; f(x)$$

$$\text{s.t.} \quad c_i(x) - y_i^2 = 0, \quad i = 1, 2, \dots, |\mathcal{I}|$$

$$c_i(x) = 0, \quad i = |\mathcal{I}|+1, \dots, |\mathcal{I}|+|\mathcal{E}|$$

$$x \in \mathbb{R}^n$$

$$y \in \mathbb{R}^{|\mathcal{I}|}$$

$$\min \; g(w) = f(x) \qquad w = [x^T \;\; y^T]^T$$

$$\text{s.t.} \quad \bar{c}_i(w) = 0, \quad i = 1, \dots, |\mathcal{I}|+|\mathcal{E}|$$

$$w \in \mathbb{R}^{n+|\mathcal{I}|}$$

$$\bar{c}_i(w) = c_i(w) - y_i^2, \quad i = 1, 2, \dots, |\mathcal{I}|$$

$$\bar{c}_i(w) = c_i(w), \quad i = |\mathcal{I}|+1, \dots, |\mathcal{E}|$$

$$\boxed{\min \; L_A(w, \lambda; \mu) = g(w) - \lambda^T \bar{c}(w) + \frac{1}{2}\mu \|\bar{c}(w)\|^2}$$

Example

$$\min \; g(w) = f(x)$$

$$\text{s.t.} \quad \bar{c}_1(w) = c_1(x) - y_1^2 = 0$$

$$\bar{c}_2(w) = c_2(x) - y_2^2 = 0$$

$$\bar{c}_3(w) = c_3(x) = 0$$

$$\min \; L_A(w, \lambda; \mu) = g(w) - \lambda^T \bar{c}(w) + \frac{1}{2}\mu \|\bar{c}(w)\|^2$$

$$\nabla_w L_A(w, \lambda; \mu) = \nabla g(w) - \lambda^T \nabla \bar{c}(w) + \mu\, \bar{c}(w)^T \nabla \bar{c}(w)$$

$$= \nabla g(w) + (\mu \bar{c}(w) - \lambda^T)\, \nabla \bar{c}(w)$$

$$= \begin{bmatrix} \nabla f(x) \\ 0 \\ 0 \end{bmatrix} + \left(\mu \begin{bmatrix} c_1(x) - y_1^2 & c_2(x) - y_2^2 & c_3(x) \end{bmatrix} - \begin{bmatrix} \lambda_1 & \lambda_2 & \lambda_3 \end{bmatrix} \right)$$

$$\times \begin{bmatrix} \nabla c_1(x) & \nabla c_2(x) & \nabla c_3(x) \\ -2y_1 & 0 & 0 \\ 0 & -2y_2 & 0 \end{bmatrix}$$

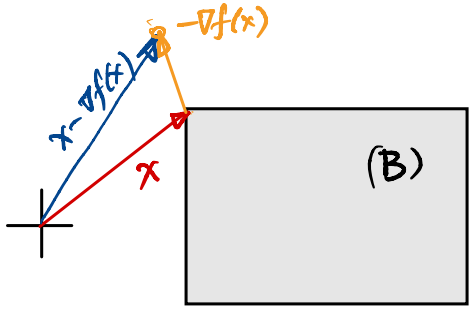
Generally:

$$\nabla_w L_A = \begin{bmatrix} \nabla f(x) \\ 0 \end{bmatrix} - \begin{bmatrix} \nabla c_I(x) & \nabla c_E(x) \\ -2Y & 0 \end{bmatrix} \left(\lambda - \mu \begin{bmatrix} c_I(x) - y^2 \\ c_E(x) \end{bmatrix} \right)$$

$$Y = \operatorname{diag}(y), \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}, \quad y^2 = \begin{bmatrix} y_1^2 \\ \vdots \\ y_m^2 \end{bmatrix}$$

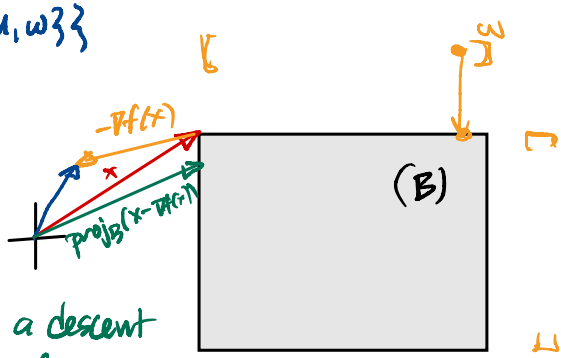
Suppose that the only inequality constraints are box constraints $\ell \le x \le u$.

Recall the optimality test:
If $-\nabla f(x) \in N(x)$ then x is locally optimal

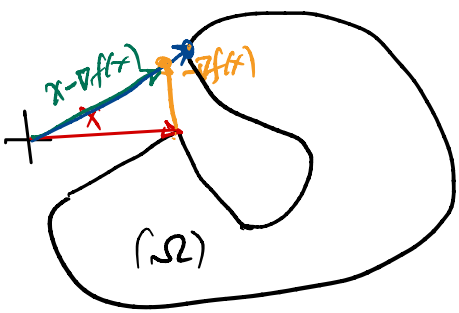


$x = \mathrm{proj}_B(x - \nabla f(x))$
$\iff$ x is locally optimal

$\mathrm{Proj}_B(w) = \max\{\ell, \min\{u, w\}\}$



$y - x$ is a descent direction from x

$y = \mathrm{proj}_B(x - \nabla f(x)) \ne x$

Procedure: Modify the unconstrained algorithm to project x_k onto (B) at each iteration.